Whisper Transcription Workflow

This lab covered prompted transcription, unprompted transcription, chunking, and export of timestamped results using Whisper. The main goal was to see how context and segmentation affect transcription quality and how the results can be saved in useful formats.

The prompted and unprompted transcriptions were different in a few ways. The prompted version was better at recognizing context-specific language, including names, topic-related words, and dialect forms. In particular, the prompt helped with words like “Aunt Blue,” “fixin’ to,” “workin’,” “throwed,” and “feller.” The unprompted version was understandable, but it made more mistakes and was more likely to normalize spoken dialect into more standard language. Even with prompting, some errors remained, especially around phrases like “short fellow/man,” which the model still misunderstood in parts of the transcript. This showed that prompting improves accuracy, but it does not guarantee perfect results.

Chunking was useful because it made the audio easier to process and organize. By splitting the recording into 30-second chunks (that turned to less via Whisper classification of understandable parts), each piece could be transcribed separately instead of sending one long file at once. This made the workflow easier to debug and easier to manage. Chunking also made timestamping possible in a structured way, because each chunk could be transcribed with its own local timestamps and then adjusted by adding the chunk offset back into the full timeline. That allowed the final transcript to match the original audio more accurately.

Several challenges came up during the lab. First, some audio libraries were missing at the start, so they had to be installed before the notebook could load and inspect audio properly. Second, chunking was more difficult than expected because the notebook environment did not initially recognize `ffmpeg` and `ffprobe`, even after installation. That caused errors until the environment path issue was understood. Another challenge was that the Step 3 transcription only needed a short sample, so adding extra clipping logic inside the notebook was unnecessary and created avoidable problems. The last challenge was making sure the timestamps were adjusted correctly after chunking so the final transcript stayed aligned with the original recording.

To improve accuracy, I would recommend using short, focused prompts with names, technical terms, and important context words. I would also recommend chunking long recordings with a small overlap between chunks to reduce the chance of cutting off words at boundaries.